

NORMAL NEWBORN

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DEFINITIONS :

1. Newborn period : First 4 weeks — { 1st week : Early.
Next 3 weeks : Late.

2. Birth weight :

- Low birth weight (LBW) : < 2.5 Kg.
- Very Low Birth Weight (VLBW) : < 1.5 Kg
- Extremely Low Birth Weight (ELBW) : < 1 Kg

Intrauterine Growth Retardation

00:05:04

SGA — { Constitutional (Normal)
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Pathological (IUGR)

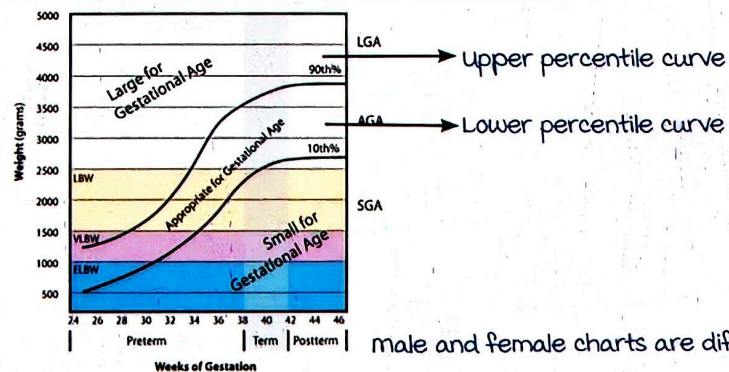
Identification points :

- Loose skin folds.
- Skin peeling.
- Thin umbilical cord.

	Symmetrical IUGR	Asymmetrical IUGR
Etiology	maternal	Fetal
Cause	uteroplacental insufficiency	Genetic defects (Eg : Trisomy), organ anomalies
Onset	2 nd & 3 rd trimester (During placenta-dependent growth of body)	1 st trimester
Effect	Brain sparing effect present ↓ Normal size head, body size small for gestational age	No brain sparing effect ↓ uniformly small baby
Outcome	Good	Poor

Assessment of IUGR :

1. New Allms intrauterine growth curves :



a. Ponderal index (PI) :

$$= \frac{\text{Weight of baby (g)}}{[\text{Length of baby (cm)}]^3} \times 100$$

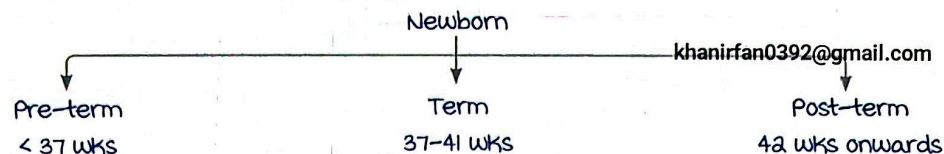
Interpretation :

- > 2 : Symmetrical IUGR.
- < 2 : Asymmetrical IUGR.

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Preterm Baby

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**ANTENATAL ASSESSMENT :**

Based on :

- mother's last menstrual period (Lmp).
- 1st trimester ultrasound scan.

POST NATAL ASSESSMENT :**Expanded New Ballard Score (ENBS) :**

- Based on — Neuro muscular maturity
Physical maturity
- Total score range : -10 to +50.
-10 : 20 weeks.
+50 : 44 weeks.
- Accuracy : ± 1 week.

Neuromuscular Maturity

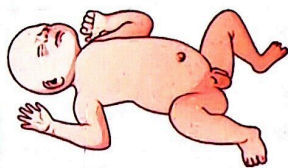
Score	-1	0	1	2	3	4	5
Posture							
Square window (wrist)							
Arm recoil							
Popliteal angle							
Scarf sign							
Heel to ear							



Pre-term baby in extended position

Physical Maturity

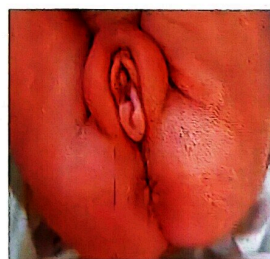
Skin	Sticky, friable, transparent	Gelatinous, red, translucent	Smooth, pink; visible veins	Superficial peeling and/or rash; few veins	Cracking, pale areas; rare veins	Parchment, deep cracking; no vessels	Leathery, cracked, wrinkled
Lanugo	None	Sparse	Abundant	Thinning	Bald areas	Mostly bald	Maturity Rating
Plantar surface	Heel-toe 40-50 mm: -1 <40 mm: -2	>50 mm, no crease	Faint red marks	Anterior transverse crease only	Creases anterior 2/3	Creases over entire sole	Score Weeks
Breast	Imperceptible	Barely perceptible	Flat areola, no bud	Stippled areola, 1-2 mm bud	Raised areola, 3-4 mm bud	Full areola, 5-10 mm bud	-10 20
Eye/Ear	Lids fused loosely: -1 tightly: -2	Lids open; pinna flat; stays folded	Slightly curved pinna; soft; slow recoil	Well curved pinna; soft but ready recoil	Formed and firm, instant recoil	Thick cartilage, ear stiff	-5 22
Genitals (male)	Scrotum flat, smooth	Scrotum empty, faint rugae	Testes in upper canal, rare rugae	Testes descending, few rugae	Testes down, good rugae	Testes pendulous, deep rugae	0 24
Genitals (female)	Clitoris prominent, labia flat	Clitoris prominent, small labia minora	Clitoris prominent, enlarging minora	Majora and minora equally prominent	Majora large, minora small	Majora cover clitoris and minora	5 26
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							35 38
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							45 42
							50 44



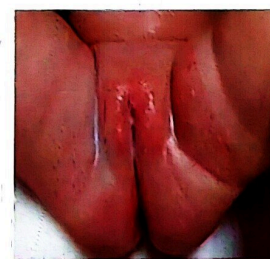
Term baby in flexed position



Scarf sign positive



(a)



(b)

a). Preterm female external genitalia

b). Term female external genitalia

Normal Newborn

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DIFFERENCES BETWEEN PRETERM & TERM BABY :

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Features	Preterm baby	Term baby
1. Posture of limbs	Extended (↓ tone)	Flexed (Normal tone)
2. Square window test	Fingers do not touch forearm (↓ flexibility)	Finger touches forearm (Flexible wrist)
3. Arm recoil	Absent	Present
4. Scarf sign	Elbow crosses midline (D/t ↓ tone & resistance)	Elbow does not cross midline
5. Popliteal angle (Angle b/w thigh & leg when flexed)	Obtuse	Acute
6. Heel to ear	Possible	Not possible
7. Breast buds	< 5 mm	> 5 mm
8. External genitalia- Female	Both labia majora & labia minora seen	Only labia majora seen (Labia minora hidden)
9. External genitalia-male	Scrotum : Less pigmented Empty scrotum d/t incomplete descent of testis	Scrotum : more pigmented Scrotum contains testis (Fully descended by 36 wks)
10. Lanugo hair	Abundant	Less/absent
11. Ear pinna recoil	Slow (D/t poorly developed elastic cartilage)	Fast

COMPLICATIONS OF PRETERM BIRTH :

Preterm complications

D/t organ immaturity :

Respiratory distress syndrome (RDS) : D/t surfactant deficiency.

- Necrotizing enterocolitis (NEC).

- Intraventricular hemorrhage : D/t bleeding from germinal matrix into ventricles (Fragile blood vessels).



- Kernicterus :

Bilirubin crosses immature blood brain barrier (BBB).

Brain damage

D/t decreased reserve of nutrients :

- ↑ risk of hypoglycemia.
- ↑ risk of anaemia (D/t ↓ iron stores).
- ↑ risk of hypothermia (D/t ↓ defence mechanisms to cold).

Normal Observations in a Newborn

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Sign	0	1	2
Activity (muscle tone)	Limp (Absent tone)	Some flexion	Good flexion/active movements
Pulse (Heart rate)	Nil	<100/min	>100/min
Grimace (Reflex irritability)	No response	Grimace, flaccid response without cry	Cry/cough or sneeze
Appearance (Colour)	Blue or pale	Body pink Extremities blue : Acrocyanosis	Pink all over
Respiratory effort	Nil	Irregular, slow breathing	Good cry

APGAR scoring :

Timing : 1 min after birth

↓ Followed by

Total maximum score : 10.

Total minimum score : 0.

Normal APGAR score : 7 to 10/10.

Low APGAR score : < 7/10.

Feedback

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Note :

- Nasogastric tube is used to irritate nasal mucosa to elicit grimace.
- Normal newborn heart rate : 120-160/min.

VITAL SIGNS :

- Heart rate : > 100/min { Term : 100-160/min
Preterm : 120-180 min
- Respiratory rate : 40-60/min.
- Capillary refill time (CRT) :

Checked by pressing ball of finger over sternum

Blanching of skin

Skin becomes pink

CRT ≤ 3 sec

- Temperature : 36.5 °C to 37.5 °C.

NORMAL FINDINGS :

Findings	Details
1. Erythema toxicum/erythema neonatorum	 → Papulosaquamous lesions 648c85cf68b03674b182fab - Histology : Eosinophils present. - Onset : > 24 hr after birth
2. Pustular melanosis	• Pustules + hyperpigmented patches. • Onset : < 24 hr after birth. • Histology : Neutrophils present.
3. Epithelial inclusion cyst	 → Keratin deposits (milia) Other sites of keratin deposition : • Palate : Epstein pearls. • Gum : Bohn's nodules.
4. Mongolian spots	 → Hyperpigmented macules D/t presence of melanocytes in dermis instead of epidermis Pathogenesis : Arrest in melanocyte migration from neural crest.
5. Salmon patch/stork bite/birth mark	 → Salmon-colored patch on forehead (Or nape of neck). D/t vascular ectasia (Dilation).

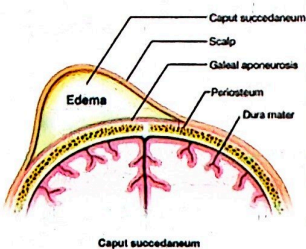
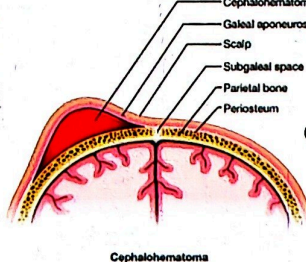
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Findings	Details
6. Breast findings	- Breast engorgement. Witch's milk: milk-like fluid discharge from nipple.
7. Bleeding per vagina	uterine shedding in female babies → sudden withdrawal of maternal hormones.
8. Small subconjunctival/retinal hemorrhage	• D/t stress of normal delivery
9. Hymenal skin tags	In female baby
10. Physiological phimosis	In male baby

Note : These conditions do not require treatment, **only reassurance**.

Head Swellings in Newborn

00:57:58

	Caput succedaneum	Cephalhematoma	Sub galeal hemorrhage
Incidence/severity	more common	Less common	more severe
Extent	Diffuse swelling	Localised swelling	Diffuse swelling
Plan	Superficial	Deep	Superficial
Location	Scalp	Sub-pereosteal	Below galea aponeurotica
Reason	D/t prolonged labour ↓ venous congestion ↓ Fluid leakage	D/t trauma to skull in instrumental delivery	Vacuum extraction ↓ Traction on layers of scalp ↓ Separation of scalp layers ↓ Blood accumulation
Content	Fluid	Blood	Blood (Fluctuant swelling)
Onset	At birth/soon after birth	Slowly increased after 12-24 hr after birth	Appears soon after birth
Recovery	2-3 days after birth	3-8 wks after birth	-
Associations	None 	- Linear skull fractures. - Jaundice (D/t heme breakdown). 	- Shock (D/t ↓ blood volume). - Pallor. - Jaundice. 